

Bayesian and Frequentist Semantics for Common Variations of Differential Privacy: Applications to the 2020 Census

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Abstract

The purpose of this paper is to guide interpretation of the semantic privacy guarantees for some of the major variations of differential privacy, which include pure, approximate, Rényi, zero-concentrated, and f differential privacy. We interpret privacy-loss accounting parameters, frequentist semantics, and Bayesian semantics (including new results). The driving application is the interpretation of the confidentiality protections for the 2020 Census Public Law 94-171 Redistricting Data Summary File released August 12, 2021, which, for the first time, were produced with formal privacy guarantees.

1 Introduction

Differential privacy [29] is the formal privacy framework adopted for disclosure avoidance in the 2020 United States Decennial Census of Population and Housing. Officially invented in 2006, it allows for rigorous mathematical reasoning about confidentiality protections and utility of privacy-protected data. Initially, the term *differential privacy* referred to a single privacy definition [29] (now known as *pure differential privacy*) but it has evolved into a family of related privacy definitions, each with its own parameters and interpretations.

The purpose of this paper is to introduce, to a statistically-oriented audience, the major variations of differential privacy that are particularly relevant to the disclosure avoidance system for the 2020 Census. These are pure differential privacy [29], approximate differential privacy [28], Rényi differential privacy [51], zero-concentrated differential privacy [14], and f -differential privacy [24].¹ We focus primarily on the privacy semantics of these definitions and consider several different viewpoints:

- *Parameter interpretability*: Each privacy definition has its own set of privacy-loss parameters, such as the ϵ in pure differential privacy. Since privacy-loss parameters must be chosen by policymakers, it is important to gain a comprehensible, minimally technical understanding of the parameters that can be used in policy discussions. Thus, where possible, we discuss the interpretations of the different parameters. Not all parameters have minimally technical interpretations, however, and, in particular, the δ parameter of the approximate differential privacy framework is often misinterpreted. We also propose a more interpretable version of approximate differential privacy, which we call *probabilistically-bounded differential privacy*.
- *Frequentist hypothesis testing*: All variations of differential privacy covered in this paper are based on a concept called the *privacy-loss random variable* (PLRV). The PLRV has close connections to

¹For other variations of differential privacy, the interested reader can consult the survey article by Desfontaines and Pejó [22].

the likelihood-ratio test and, as a consequence, all these definitions provide protections that can be framed as frequentist hypothesis tests. If an attacker is using the privacy-protected data to perform a hypothesis test related to the confidential information supplied by an individual, the relationship between the significance level and power of *any* such test is governed by the privacy-loss parameters of the variation's definitions.

- *Bayesian semantics:* There are fewer results related to Bayesian semantics owing to the mathematical complexity of analyzing Bayes' rule when differentially private randomization is combined with an attacker's prior information. We explain the Bayesian semantics of pure differential privacy and introduce new results for Rényi and zero-concentrated differential privacy. These results have close connections to the [probabilistically-bounded differential privacy](#) definition we introduce in this paper. Traditional Bayesian disclosure avoidance methodology compares what can be learned about a target individual based on a data release to what would have been known without the data release (also known as a *prior-to-posterior* comparison). This comparison is problematic because *anything* learned from a data release (including generalizable knowledge such as smoking causes cancer) is treated as a privacy violation. The literature on differential privacy examines the more nuanced question of what can be learned about an individual based on a data release compared to what could have been learned from the data release had the individual's information been scrubbed beforehand. This comparison is known as a *posterior-to-posterior* comparison and avoids the pitfalls of prior-to-posterior comparisons. Thus the Bayesian semantics we discuss are of the posterior-to-posterior variety.

When discussing these semantics, we also apply them to the production settings of the 2020 Census Disclosure Avoidance System (DAS). We show that different privacy semantics can be fine-grained and tailored to the types of inferences an attacker may try to make. This stands in contrast to the common mistaken belief that only one coarse-grained guarantee, summarized by a single number, is possible.

One might ask why this paper covers six variations of differential privacy. Isn't one enough? It turns out that if we want comprehensive privacy guarantees from the viewpoints of parameter interpretability as well as frequentist and Bayesian statistics, then all six variations are needed. Specifically, pure ϵ -differential privacy [29], the ancestor of all these variations, is the starting point. It provides the strongest privacy protections and is easy to analyze from both the frequentist and Bayesian points of view. In certain cases, it may over-estimate the privacy risks that individuals face even against extremely powerful attackers. Hence, approximate (ϵ, δ) -differential privacy was introduced to better track those privacy risks. Approximate differential privacy represents a continuum of privacy definitions – an (ϵ, δ) -curve [9, 24] – although many articles unfortunately consider just a single point on that curve (like trying to understand an elephant by studying a single hair on its tail). Approximate differential privacy is easy to study from the frequentist point of view [42], but Bayesian results are more limited [43]. Furthermore, the δ parameter is not directly interpretable and is often misinterpreted. To address this issue, we propose (ϵ, δ) -[probabilistically-bounded differential privacy](#), which has more interpretable privacy-loss parameters. When it is treated as a continuum of privacy definitions (i.e., an (ϵ, δ) -curve), [pbdp](#) is a reparametrization of f -differential privacy (f -DP) [24] implying that f -DP is a natural consequence of addressing the difficulties in interpreting the δ parameter of approximate differential privacy. In addition to parameter interpretability, f -DP also has natural frequentist semantics because it was specifically developed to manage the trade-off between the level and power of re-identification inferences [24]. Except in special cases, the privacy parameters of f -DP (and [probabilistically-bounded differential privacy](#)) are difficult to compute, but Rényi and zero-concentrated differential privacy can be used to compute upper bounds on those parameters and on the frequentist semantics of f -DP and [probabilistically-bounded differential privacy](#). Furthermore, we propose new Bayesian privacy semantics for Rényi and zero-concentrated differential privacy that, except in special cases, are difficult to obtain when working directly with f -DP and [probabilistically-bounded differential privacy](#). Unfortunately, however, the privacy parameters of Rényi and zero-concentrated differential privacy are not easy to interpret. To summarize, parameter and frequentist interpretability is a strength of f -DP and [pbdp](#) while Bayesian semantic interpretability and ease of parameter computation are strengths of Rényi and zero-concentrated DP.

We note that there is a natural trade-off between privacy and utility in a data release. It is up to the data collector (e.g., statistical agency) to carefully balance the privacy and utility requirements of their applications. The utility side is dataset-specific, but privacy can be treated in a more general way. Thus,

this paper discusses various ways in which privacy protections can be quantified in order for them to be used in privacy/utility discussions.

This paper is organized as follows. In Section 2, we provide an accessible and informal introduction to what is and isn't considered a confidentiality breach in modern statistical disclosure limitation (SDL). Such a discussion is especially important given that many (peer-reviewed) articles about privacy and confidentiality are based on intuitive but faulty premises – faulty in the sense that the natural conclusions of those premises are contradictory. In Section 3, we introduce the notation used in this paper. In Section 4, we discuss the privacy-loss random variable (PLRV), which is the technical underpinning of all the privacy definitions we consider and has close connections to the likelihood ratio test. We also discuss the desirable technical properties that a “good” formal privacy definition should have. In Section 5, we explain all the privacy definitions, propose *probabilistically-bounded differential privacy*, and discuss the interpretability of each definition's parameters. Frequentist privacy semantics are discussed in Section 6 and Bayesian semantics are discussed in Section 7, including new semantics for Rényi and zero-concentrated differential privacy. Privacy analyses can be fine-grained, depending on the information an attacker may wish to infer about a person potentially represented in the confidential data. These fine-grained attacks, with applications to the 2020 Census, are discussed in Section 8. Finally, we discuss additional related work in Section 9 and conclude in Section 10.

2 Informal Introduction to Privacy Semantics

Privacy and confidentiality are two different concepts. Informally, privacy relates to a person's ability to control access to information about themselves and confidentiality refers to a third-party's (e.g., a statistical agency's) ability to prevent disclosure of information it has collected. However, in the computer science literature, the word “privacy” is frequently used to mean “confidentiality” (as in the case of differential privacy) and that is also the way the term privacy will be used in this paper.

With this caveat about terminology, we note that there is a common belief that a privacy breach occurs whenever a public dataset (derived from confidential data) is used to make unwanted or potentially harmful inferences about an individual. This belief is quite wrong, as it encompasses situations that are not privacy breaches. One important case is when the unwanted inference depends on generalizable (statistical) knowledge contained in the public dataset rather than on the specific datum that a target individual may or may not have contributed to the underlying confidential database. It is difficult to over-emphasize this distinction, and we take pains here to make it as clear as possible.

Thus, consider the canonical example of the first Cancer Prevention Study, also known as CPS-I [39, 6], discussed by Dwork and Pottenger [26]. This large-scale study, which followed a cohort of volunteers from 1952 to 1972 was the first study to conclusively establish the link between smoking and cancer. Based on the scientific knowledge gained in this study, if you see a Millennial chain smoker, you can be virtually certain that they have a much higher chance of developing lung cancer than if they never smoked. Such an inference is definitely unwanted, as it can result in higher health and life insurance premiums. However, for people born after 1972, like our Millennial, this study cannot possibly be considered a privacy breach because their data could not have been used in the study. For such people, one would say that the inference is purely statistical in nature.

Therefore, an unwanted inference is only a privacy breach if it is specifically *caused* by the inclusion of the individual's information in the dataset from which the inference was made. Consequently, an empirical privacy analysis needs to distinguish between the causal effect on inference due to a particular individual's data being used and the general statistical information provided by the dataset on a group of other individuals. A privacy-protection methodology must work to limit the causal inference about a particular individual (to protect privacy) while leaving intact statistical inferences due to information in the data, which is basically the same as leaving the data fit for statistical purposes [61].

To relate the causal inference interpretation of a privacy breach to concepts that have been used historically, consider the textbook definitions of identity and attribute disclosure. In traditional SDL, an *identity disclosure* occurs when “a data subject is identified from released data” [25, p. 174], an *attribute disclosure* occurs when “information [is disclosed] about a population unit without (necessarily) the identification of the unit within the data set” [25, p. 172], and an *inferential disclosure* occurs when either an identity or

attribute disclosure can be inferred with high probability [18]. The traditional SDL literature has not been careful to distinguish between identity and attribute inferences that depend upon the use of the respondent's data and those that are possible without using the individual's information. The confidentiality-breaching versions of identity, attribute and inferential disclosure all have the same mathematical formulation, which we now illustrate.

The differential privacy framework for SDL methods is designed precisely for the purpose of distinguishing between confidentiality breaches and valid scientific inferences. Typically, one reasons about causality by studying *interventions* [55], which are direct manipulations of a system. In the case of differential privacy, the intervention is to replace an individual's data record with something else and to reason about the effect of this intervention [61].

More concretely, suppose *Jessie* is a 52-year-old Hispanic Asian woman and let her survey response (e.g., to the decennial census) be denoted by r . Let $\mathcal{D}$ be the collection of responses of everyone else, so that $\mathcal{D} \cup \{r\}$ is the complete dataset (*Jessie*'s record combined with everyone else's records). Let M be an algorithm that applies SDL protections while producing an output ω . Mathematically, $\omega = M(\mathcal{D} \cup \{r\})$. Both ω and the source code of the mechanism M are then made public. One may wonder what does ω (and knowledge of M) reveal about *Jessie*, and how much of this is due to their record being part of the input to M .

Following the causal view [61], one might ask what would have happened in an alternate world in which the data collector (e.g., the Census Bureau) replaced *Jessie*'s submitted record r with some other fixed, pre-selected record r^* (e.g., a 60-year-old White Non-Hispanic male) before running M . In such a world, the output would be denoted as ω^* . Mathematically, $\omega^* = M(\mathcal{D} \cup \{r^*\})$. Clearly, the specific contents of the record r that *Jessie* submitted can have no causal effect on ω^* since r was never used. Therefore, ω^* has no causal effect on inferences about *Jessie*'s recorded data. All inferences about *Jessie* based on ω^* are statistical inferences (e.g., based on the demographic composition of *Jessie*'s community), not privacy-breaching inferences about how *Jessie* differs from their community.

One could then compare inference about *Jessie* in the actual world, where the output of $M(\mathcal{D} \cup \{r\})$ is observed, to inference about *Jessie* in her *privacy-preserving counterfactual world*, in which $M(\mathcal{D} \cup \{r^*\})$ is observed. The way this difference is measured is at the heart of the different commonly-accepted variations of differential privacy [29, 28, 14, 51, 24].

We briefly note that the degree to which an inference is unwanted (i.e., potentially harmful to the particular individual or entity) is also an important consideration. For example, a healthy individual may have fewer concerns about revealing some medical data than an individual who is not. It is possible to add specifications about which types of inference are unwanted in formal privacy definitions (e.g., [47, 41, 48, 35]). However, for simplicity of the explanations, in this document we consider the case where all such inferences are considered potentially sensitive, which is consistent with the statutory framework regulating the Census Bureau and other statistical agencies in the United States (13 U.S. Code §§ 8(b) & 9, 44 U.S. Code §§ 3561(11) & 3563). This also allows individuals to retroactively change their mind about which inferences they would consider harmful.

Now, comparing inferences about *Jessie* when her record is used (i.e., based on $M(\mathcal{D} \cup \{r\})$) to when her record is not (i.e., $M(\mathcal{D} \cup \{r^*\})$) is also tricky. To see why, suppose M simply outputs the number of 52-year-old females in the input data and suppose, with *Jessie*'s record, there would be 523 of them (i.e., $M(\mathcal{D} \cup \{r\}) = 523$). In *Jessie*'s privacy-preserving counterfactual world, the answer would be 522 (i.e., $M(\mathcal{D} \cup \{r^*\}) = 522$). How does inference about *Jessie*'s age compare to the inference in the counterfactual world? In the counterfactual world, nothing is revealed about *Jessie*'s age. However, in the actual world, it is not clear what the answer 523 reveals – it all depends on what an attacker knows, what additional information is available to the attacker, and how the attacker chooses to perform the statistical inference. There is no general agreement about reasonable choices here – in any room of 10 economists, statisticians, social scientists, or demographers, there would be at least 23 different, and often mutually contradictory, suggestions.

Differential privacy provides a clever way of sidestepping the question of what does 523 reveal about *Jessie* compared to 522. The main idea is to force M to be randomized: sometimes $M(\mathcal{D} \cup \{r\})$ will produce 523, sometimes it will produce 522, and other times it will produce other numbers. Similarly, sometimes $M(\mathcal{D} \cup \{r^*\})$ will produce 523, sometimes it will produce 522, etc. Thus instead of reasoning about 522 vs. 523, differential privacy reasons about how likely you are to see 523 if the input is $\mathcal{D} \cup \{r\}$ compared to when the input is $\mathcal{D} \cup \{r^*\}$. If the output distribution, when $\mathcal{D} \cup \{r\}$ is the input, is exactly the same as when